B.Tech. Comp. Engg. Vth Sem Examination 2025
AUTOMATA THEORY
Paper no CSC-505/CEN-502 (Old)

Max Time : Three hours **Max. Marks: 45/60**

Attempt any TWO parts from each question.
Assume missing data if any.

Write your Roll No here ..23BCS066..... on receipt of the question paper immediately

Q. No	CO / BL	Questions	Marks
Q1 a)	1 / L2	Explain the role of state diagrams and state tables in representing finite automata. How does each representation help in understanding the behavior of the automaton?	4.5/6
Q1 b)	1 / L3	Given an NDFA with states {q0, q1, q2}, alphabet {0,1}, and transitions: • $\delta(q0,0) = \{q0, q1\}$ • $\delta(q0,1) = \{q0\}$ • $\delta(q1,1) = \{q2\}$ • $\delta(q2,0) = \{q2, q1\}$ *Where, q0 is the initial state and q2 is the final state.* Convert this NDFA to its equivalent DFA and provide the state table.	4.5/6
Q1 c)	1 / L3	Consider the following DFA over the alphabet {0,1}, and apply minimization : *State* | *Input = 0* | *Input = 1* | *Final?* *A* | B | C | No *B* | A | D | No *C* | E | F | Yes *D* | E | F | No *E* | E | F | Yes *F* | F | F | No **Verify your answers using suitable sentences.**	4.5/6
Q2 a)	2 / L2	Simplify the regular expression **((a+b)*a(a+b)*)+((a+b)*b(a+b)*)** using regular expression identities.	4.5/6
Q2 b)	2 / L1, L5	State and Define the Pumping Lemma for Regular Languages. Also evaluate the effectiveness and limitations of the Pumping Lemma in proving non-regularity of languages.	4.5/6
Q2 c)	2 / L3	Apply this lemma to analyse the Regularity of the following language : L={w∈{a,b}*|w contains an equal no. of occurrences of the substring "ab" and "ba"}	4.5/6

Q3 a)	2 / L2, L4	Design a context-free grammar that generates the language. $L=\{a^n b^m \mid n\geq 1, m\geq 1, n=m \text{ or } n=2m\}$ **Verify your answers using suitable sentences.**	**4.5/6**
Q3 b)	3 / L3	Consider the following CFG, where {S, NP, VP, Det, N, V} is the set of variables. S → NP VP NP → Det N VP → V NP Det → the N → man | telescope V → saw Generate the LMD, RMD & Derivation tree for the sentence "***the man saw the telescope***", and also comment on the nature of the given CFG *w.r.t.* ambiguity.	**4.5/6**
Q3 c)	3 / L4	Compare and contrast the computational power and grammar restrictions of the four Chomsky language types.	**4.5/6**
Q4 a)	4 / L1, L2	Using a labelled block diagram explain the differences between finite automata (FA) and pushdown automata (PDA) in terms of memory and language recognition power. Also describe how the stack in a PDA helps in recognizing non-regular languages with an example.	**4.5/6**
Q4 b)	4 / L4	Given the PDA transitions: **δ(q0, a, Z0) = (q0, aZ0)** **δ(q0, a, a) = (q0, aa)** **δ(q0, b, a) = (q1, ε)** **δ(q1, b, a) = (q1, ε)** **δ(q1, ε, Z0) = (qf, Z0)** Analyze the language accepted and whether it accepts by **final state** or **empty stack.**	**4.5/6**
Q4 c)	4 / L3, L4	Convert the given CFG into an equivalent PDA, showing all Steps involved & Rules applied using the standard CFG-to-PDA algorithm. **S → aSb | bSa | ε** Highlight the merits and Demerits of this method considering the above CFG.	**4.5/6**
Q5 a)	5 / L2	Explain the architecture, components & Salient features of a Turing Machine. Analyse how does it differ from a Pushdown Automaton in terms of computational power?	**4.5/6**
Q5 b)	5 / L6	Design a TM that accepts the language $L=\{a^n b^n c^n \mid n\geq 1\}$. **Verify your answer with suitable examples.**	**4.5/6**
Q5 c)	5 / L4, L3	Analyze the differences between recursive, recursively enumerable, and non-recursively enumerable languages. How are these related to Turing Machines?	**4.5/6**